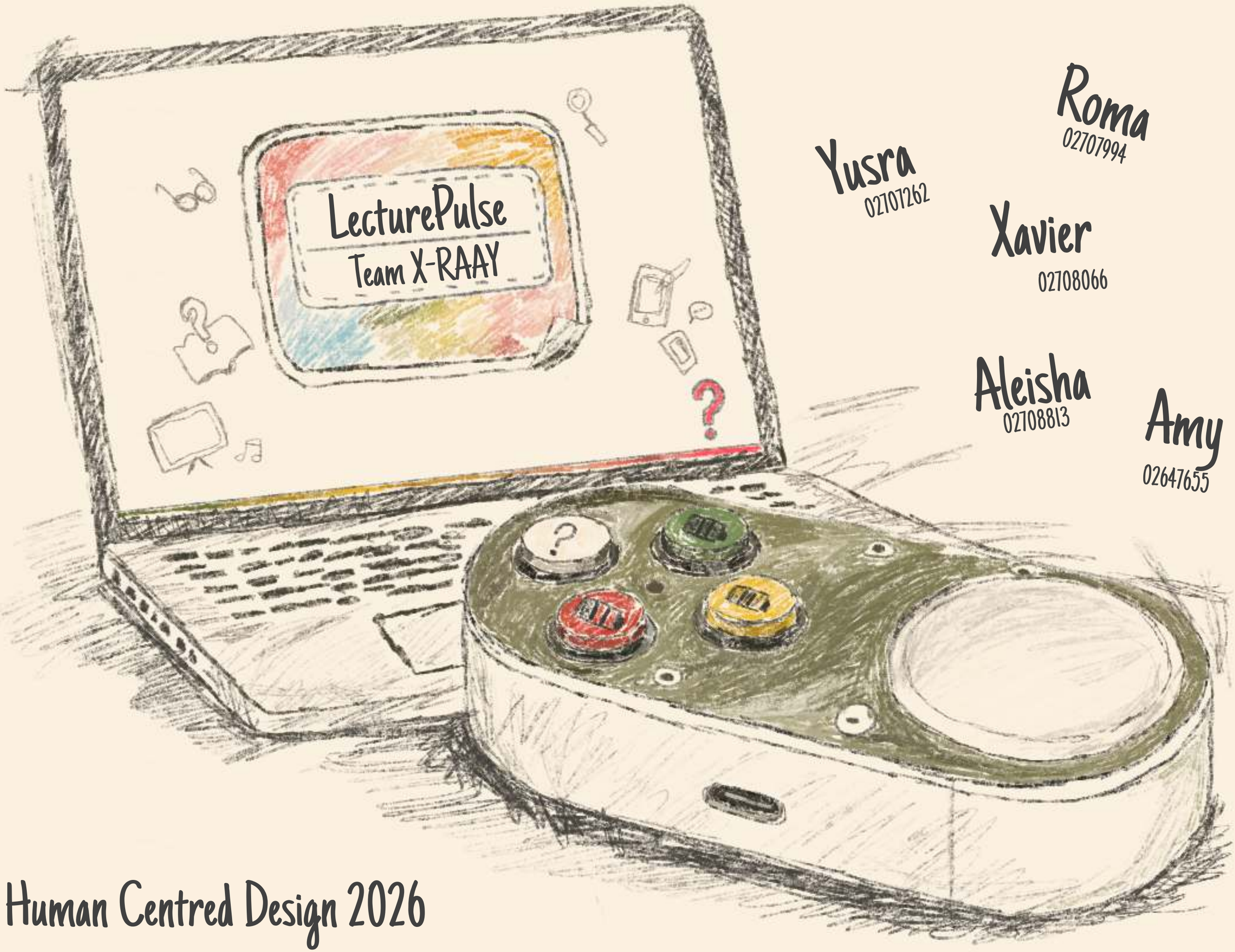


Yusra
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Human Centred Design 2026

Executive Summary

User feedback helped us narrow three concepts into a single, validated solution: the Lecture Reaction Pad.

HMW help lecturers maintain interactive lectures while reducing cognitive workload and supporting sustained student engagement?

Chosen concept

Lecture reaction pad

Problem:
Limited student–lecturer communication reduces engagement and awareness of student understanding.

Why:
The Lecture Reaction Pad enables quick, real-time communication to support more responsive and engaging lectures.

Pre-concept selection – Primary Users & Stakeholders

University students were identified as the primary users and lecturers recognised as an additional key stakeholder. The 3 concepts aim to support engagement while enhancing the lecture experience.

Prototyping approach

We used a student-led co-design approach, supported by lecturer feedback and iterative user testing throughout the design process, from low-fidelity concepts to high-fidelity prototypes.

Interviews and insights

We conducted interviews with both students and lecturers to understand key needs within the lecture environment, from which we identified themes around motivation and engagement, personal growth, supportive learning environments and lecture pacing and energy. Students highlighted the need for simple ways to stay engaged and reflect on understanding, while lecturers emphasised the importance of clear, real-time signals to adjust delivery and maintain effective lecture flow.

Next steps

Develop a working prototype of the Lecture Reaction Pad that improves real-time communication between students and lecturers during lectures.

Insights to Concepts

Project direction shaped by user challenges, insights and evolving design concepts

INSIGHTS	Motivation and engagement	Personal improvement / growth	Supportive learning environment	Lecture pacing and energy
WISH	Students wish for lectures that maintain their attention and encourage participation	Students wish to attend lectures to immerse themselves in their learning journey and make the most of their university experience	Students wish for supportive and inclusive learning environments that respect diverse needs and encourage confident engagement	Students want an environment where lecturers can adjust in response to student understanding, so pacing and delivery ensures no one falls behind
BARRIER	Low energy and passive lecture formats can make it difficult for students to engage with learning	Grade-focused mindsets and peer comparison can negatively impact confidence and motivation to really learn in lectures	Fear of judgement and openness can make it difficult for students to engage confidently, particularly neurodivergent students	Limited communication of student understanding often leaves lecturers relying on subjective classroom cues to adjust teaching
HMW'S	How might we control students' varying energy levels so they can stay engaged during lectures?	How might we help lecturers keep lectures interactive without increasing their cognitive workload?	How might we create more inclusive and accommodating lecture environments that better support neurodivergent students?	

MOOD PULSE



WHAT?
The Mood Pulse is an engagement-support device that uses gentle vibration prompts to help students focus.

WHY?
By providing users with subtle vibration prompts at timed intervals, Mood Pulse offers an intuitive way to regain focus when attention begins to drift. These pulses act as sensory cues that encourage users to re-engage with the lecture or task without causing disruption. This gives users the ability to stay connected to learning activities.

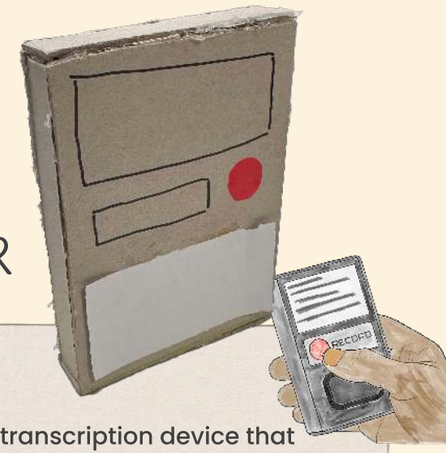
LECTURE REACTION PAD



WHAT?
The Lecture Reaction Pad is a real-time feedback tool enabling students to communicate their understanding and engagement during lectures.

WHY?
This gives students a simple way to communicate with their lecturer during lectures. This reduces the need for lecturers to rely on reading the room and gives them clearer insight into student needs, helping them deliver a more responsive learning experience.

WALK-n-TALK RECORDER



WHAT?
The Walk & Talk Recorder is an AI transcription device that records and organises lectures into clear, time-based sessions to support memory and easy retrieval of information.

WHY?
This gives students an external way to capture and organise lecture content in real time. This reduces the pressure on working memory and recall, particularly helping neurodivergent

Concept selection

User testing on lo-fi prototypes and feedback

Questions

1. What are your overall impressions of this concept?
2. Would you use this in a lecture setting? Why or why not?
3. What improvements or additional features would you suggest?

MOOD PULSE

"It doesn't really help with the lecture itself... the bigger problems are things like noise, getting confused, or just struggling to stay focused when the lecture is boring"

"The pulses can only do so much. May need more features to make a bigger difference, while still avoiding anything that could be another distraction"



We conducted co-design interviews with students as they are the key stakeholders who would use the Mood Pulse. Since the concept aims to give students greater autonomy over managing their engagement and attention during lectures, involving them in the design process helped ensure the solution reflected their needs, preferences and experiences.



POSITIVE FEEDBACK

It offers a simple way for students to self-manage attention and engagement during lectures, giving them more autonomy over their learning experience.

NEGATIVE FEEDBACK

It may have limited impact on deeper lecture issues and risks being too subtle or distracting without significantly improving focus or engagement.

LECTURE REACTION PAD

STUDENTS

"Reduces the stigma of communicating with lecturers... for fresher students and those from different cultural backgrounds"

"Layout isn't very intuitive, having the pre and during-lecture options side by side could be distracting... lecturers may not engage with the some information at all"

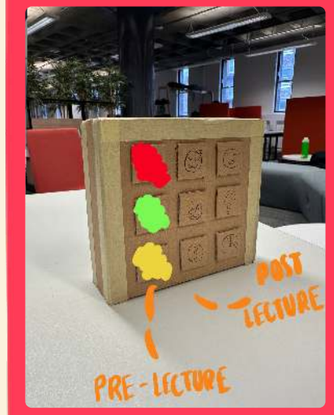


We conducted co-design interviews with both students and lecturers, as they are the two key stakeholder groups affected by the Lecture Reaction Pad. Since it relies on student feedback and lecturers responses, involving both in the design process helped us to better understand how the concept could support communication between students and lecturers in a way that is practical and beneficial for both parties.

"From a lecturer's perspective, there's no clear incentive to use this, and it could potentially add to workload rather than support it"

"If there are too many reaction options it could become overwhelming, reducing it to fewer, clearer reactions would make it more useful for quickly understanding student responses and drawing conclusions"

LECTURER
(Kamyar Hazeri)



POSITIVE FEEDBACK

It enables real-time feedback from students to lecturers, improving communication and helping lecturers better understand engagement within the lecture.

NEGATIVE FEEDBACK

It could increase lecturer workload and distraction if not carefully simplified, with concerns around usability, incentive, and information overload.

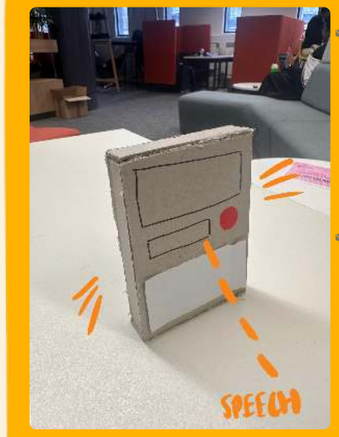
WALK-n-TALK RECORDER



We conducted co-design interviews with students as they are the primary users of the Walk-n-Talk Recorder. Since the concept is designed to support students in capturing thoughts and supporting memory during learning activities, involving them in the design process helped ensure the solution reflected their needs.

"I wouldn't use it that much personally, I tend to prefer writing things down. I can see myself using it for things like meeting or lecture recordings"

"A smaller, more portable design (pocket or keychain-sized with a carabiner) would be better. It should stay lightweight and simple to improve usability"



POSITIVE FEEDBACK

It effectively supports students by capturing thoughts and reducing working memory load, acting as a lightweight external memory tool during learning activities.

NEGATIVE FEEDBACK

It may be underused in fast-paced lecture settings and relies heavily on students actively remembering to use it in the moment.

Conclusion

Both students and lecturers agreed that the Lecture Reaction Pad has strong potential moving forward, with clear room for further innovation and impact within lecture environments. Compared to the other concepts, which do not align as well with real lecture conditions, this solution better accounts for the needs of all users and the factors that influence engagement within the lecture setting. This is why the Lecture Reaction Pad is our chosen final concept.

Development: Physical

Created five concept sketches based on the final concept decision and gathered verbal user feedback from engineering students

Attached with a Sliding Desk-edge Clip



User Feedback:

PROS
- Easy to attach and remove
- Keeps the reaction pad secure during use
- Hidden from view, reducing distractions

CONS
- Uncomfortable to sit at desk as it may hit chest
- Could become loose
- Hit the knee if under desk

Attached with Snap-Bend Action



User Feedback:

PROS
- Quick and easy to attach
- Portable and lightweight
- Can fit around different desk sizes

CONS
- Users felt it might not be secure enough
- Could wear out after repeated 'slapping'
- May accidentally detach if knocked

Inspired by slap bands

Integrated Seamlessly

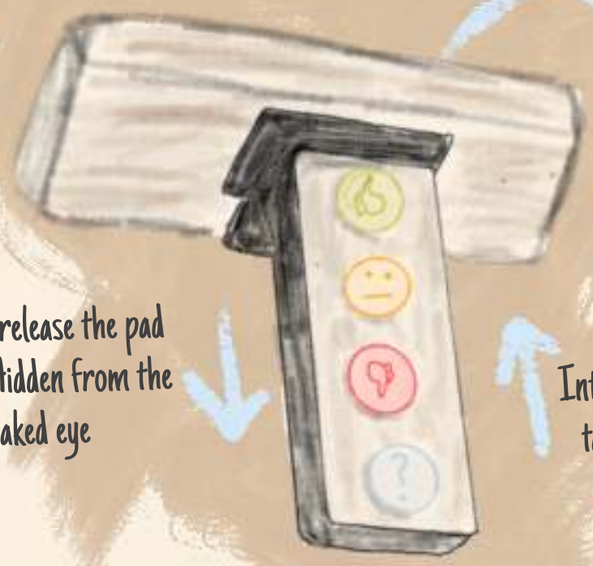


User Feedback:

PROS
- Very discreet and aesthetically pleasing
- No risk of the device falling off
- Easy for users to access during lectures

CONS
- Requires modification of existing furniture
- Expensive and difficult to install
- Not portable between classrooms

Slide In+Out Mechanism



User Feedback:

PROS
- Can be hidden when not in use
- Provides a clean and organised appearance
- Reduces clutter on the desk surface

CONS
- Moving parts could increase manufacturing complexity
- Risk of the mechanism jamming over time
- Users may forget the device is stored underneath

Chosen Design Concept



Game-like Controller

User Feedback:

PROS
- Comfortable ergonomic shape
- Familiar and engaging for students
- Buttons are easy to identify and press

CONS
- Looks more like a gaming device than an educational tool
- Could distract users during lectures
- Larger size may take up too much desk space

What was found:

Users **preferred** the **game-like controller design** because it evoked feelings of nostalgia and familiarity, **increasing their willingness to engage** with the product. Feedback suggested that students are often surrounded by lecture and study-related materials, which can **reduce motivation** and participation. By **incorporating** a design associated with leisure and positive experiences, the product becomes **more appealing** and **encourages regular use** while **maintaining its educational purpose**.

Hence this was our chosen design concept

Initial Prototypes: Functional

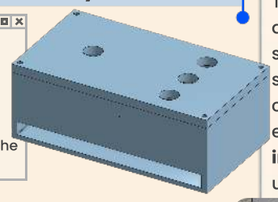
Physical 1:



Physical

One initial prototype incorporated a **slider** to represent different emotions. This feature would require a potentiometer, **increasing cost of production**. The slider introduced unnecessary complexity and could potentially become a distraction for students **from feedback**, reducing the simplicity and ease of use of the product.

Physical 2:



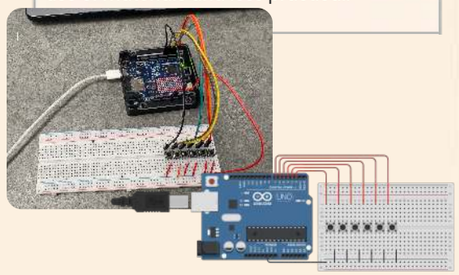
CAD Render

This led us to develop a cuboid-shaped design featuring button openings on the top surface and an **integrated phone holder** section. However, both the electronic components and the phone required the enclosure to be significantly larger, **resulting in a bulky product** that was **inefficient** in its use of materials and less practical.

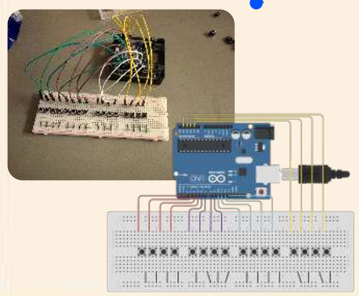
Electronics 1:

Breadboard Wiring Diagram + Real-Life

Our first electronic prototype used six buttons, with two dedicated to cycling between different students. Without multiple students, the emotion and colour aggregation feature would **not function effectively**, as it would **only be processing the feelings of a single student**. This was a very basic prototype, consisting only of a breadboard and an Arduino R4, and was created primarily **for user testing** and proof-of-concept purposes.



Electronics 2:

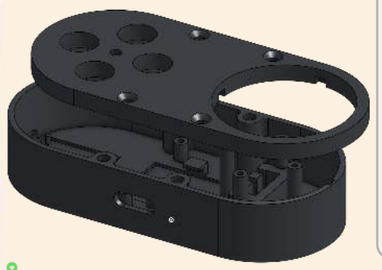


Breadboard Wiring Diagram + Real-Life

Our second electronic prototype used a larger breadboard alongside the same Arduino R4 to **more accurately simulate** how students would interact with the final product. By representing four separate students, we were **able to fully test** the emotion aggregation system without needing to cycle **between users**. This prototype also relied solely on simple push buttons and was **connected directly to a laptop via USB**. Using serial communication, the button inputs were transmitted to software running on the laptop, where the data was processed and aggregated centrally.

Leading Prototype: Functional

Physical:



Notes — Done

This led us to develop a new casing design that was **significantly smaller**, more compact, and more practical than previous iterations. Throughout the design process, we carefully considered the **placement** and **mounting of the electronic components** to ensure **easy assembly and maintenance**. Space was also allocated for **convenient USB access**, allowing the ESP module to be **programmed and charged without** requiring the device to be **disassembled**. In order to securely keep the two parts together, we used heat inserts to create our own screw holes in the casing. This, paired with counter sunk screws keep strength whilst keeping a seamless top cover.

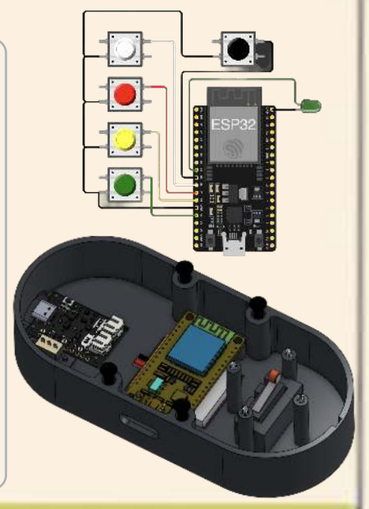
Electronics :

Notes — Done

This electronic prototype utilised an ESP32, allowing all of the **components to fit within the compact pad design**. This version incorporated the **larger user-tested buttons, along with an LED indicator and an additional large activation button**. The activation button was **designed to be pressed when a phone is placed on top** of the pad, turning on the LED and switching the device into active input mode, where button presses would contribute to the students' aggregated focus score.

Unlike previous versions, which required a constant connection to a laptop, **this design was powered independently** using a LiPo Amigo Pro and a 3.7V LiPo battery. When the switch located on the underside of the pad is turned on, power is supplied to the ESP32, which automatically connects to the network of other ESP32 devices.

The casing was designed with **integrated mounting features and dedicated spaces for each component** to ensure the electronics were securely housed for professional assembly while still allowing access for programming, and maintenance.



Software System

The software system can be broken down into a series of simple stages. Each ESP32 is programmed to transmit a dictionary containing its current inputs whenever a button is pressed. During early testing this data was sent through a Serial bridge, while the final prototype uses ESP-NOW/ Wi-Fi communication. Three ESP32 devices, representing Students 2–4, are connected wirelessly to the Student 1 ESP32, which acts as a central hub. These devices send their button inputs to the hub, which then communicates with the professor's laptop to generate the visual output displayed on screen.

Visual Interface

The visual interface was developed using Tampermonkey, a Chrome browser extension that allows custom scripts to modify and display information directly within a web page. The system assigns a numerical value to each focus input: the "Low Focus" button corresponds to -10, the "Neutral" button to 0, and the "High Focus" button to 10. Once all students have submitted their responses, the software calculates the average focus score and maps the result to a colour scale, ranging from red (low focus) to green (high focus) using RGB values.



User Feedback Addressed

Student User 1

"The simplified three-button system makes it quick to use without distracting me from the lecture."

Lecturer

"The anonymous question feature encourages participation without putting me on the spot."

Student User 2

"The colour-based display gives me useful class feedback at a glance without adding extra workload."

Development: Technical

Evaluation of various external components through user testing to determine the final design

We used CMF analysis (Colour, Material, Finish) to balance the visual, tactile and functional qualities of the product, ensuring it communicated a strong sense of value to the user. Because our primary goal was to reduce distractions and support focus, every CMF decision needed to reinforce that purpose. To apply CMF effectively, we conducted extensive user testing and explored the psychological effects of specific features. User testing during technical development was essential for ensuring the product's feel and visual appearance aligned with what users actually want and need, while also validating, or challenging, the assumptions we held as designers.

1. Colour selection

We selected a traffic-light colour system (red, yellow, green) for the buttons to represent **low, neutral, and high focus levels**, using universally understood cues for quick recognition and clear communication. [2]
The shade of red chosen was coral red (#FF4040), although highly luminous (attention capturing) [1] it's warm toned to serve as a **balance to the cool shade green** used, allowing accessibility to our buttons for **colour blind users** [2].
The shade of yellow chosen was mustard yellow (#F0D246) because of its **warm earthy tone**, it reduces visual fatigue and in addition allows the brain to release serotonin creating a **calm environment** [3].
The shade of green chosen was sage green (#9CAF88) as it is **cool toned** contrasting with the warm toned red to ensure the buttons are accessible to colour blind users [2]. This shade is also a valuable **balance 'between calming and energising'** [4].

For the main body of the product, we chose colours that support focus, as they dominate the overall visual experience. Our psychological research identified Secluded Green (#667041) as a calming, nature-associated tone that helps **regulate the nervous system** [5]. Graceful Neutral (#FFF9EB), a warm neutral, adds **softness and warmth** [5]. When used together in colour blocking, these tones create a **balanced contrast that reduces visual fatigue** and maintains a focused user experience.

CORAL RED

MUSTARD

SAGE GREEN

SECLUDED GREEN

GRACEFUL NEUTRAL

2. Icon testing

We added **icons** to the lecture reaction pad buttons to ensure **accessibility for colour-blind users**[2]. Through **A/B testing**, we evaluated different **icon** sets for the focus and question-mark **buttons**.

1 2 3



Focus icons	1&2	2&3	1&3	Total
1. Battery	10	-	12	**22**
2. Emojis	3	10	-	13
3. Thumbs	-	3	1	4

A 

B 

C 

Question mark icons	A&B	B&C	A&C	Total
A	10	-	15	**25**
B	5	13	-	18
C	-	2	0	2

Most users preferred the **battery-style icons** for the focus button, noting they **"related [more] to the idea of focus"** and linked naturally to familiar phone battery indicators. The battery icon's three levels of 'charge' also resonated with users as a **"standardised scale"**.

For the question-mark button, users favoured the **rounded style (a)**, describing it as **"easier on the eyes"**. Also saying that the rounded shape was more **"organic"**.

3. Material finish testing





Button finish	User choice
Polished shiny PLA	1
Silky matte silicone	20

We conducted **A/B testing** with users to compare the two **button finishing options**. A/B testing allowed us to **refine our design decisions** with direct feedback from users, whilst **observing** how they **interact** with the different options.

20 out of 21 participants preferred the **silky matte silicone**, describing it as "smooth" and **"more natural to touch".** Based on this strong preference, we selected a matte, silky silicone finish for the final design of the buttons.

Final Design

Overview of the final design concept accompanied by a storyboard demonstrating a realistic use case

Rendered Image of Lecture Pulse Pad



Lecture Pulse: A lecture feedback system that lets students quietly signal focus or confusion, giving lecturers a live visual pulse of the room without disrupting the flow.

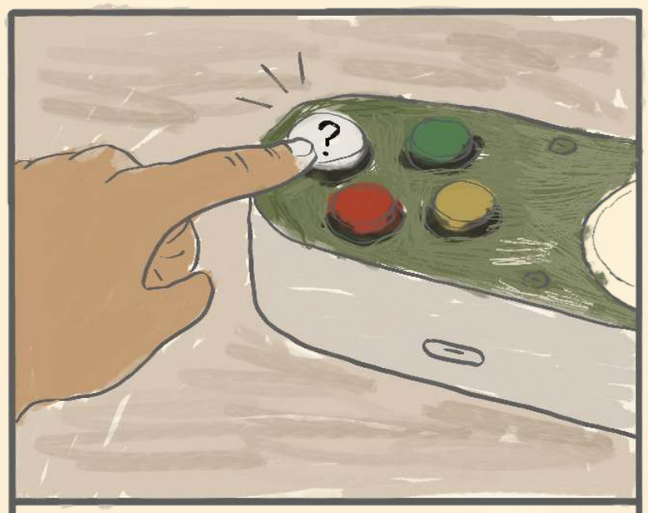


Render of Lecturer Display Visuals

Storyboard: This storyboard demonstrates a potential scenario of the **LecturePulse** in a live lecture environment.



Student struggles to understand the lecture content.



The student presses the question button for clarification.



The lecturer receives a real-time confusion alert.



The lecturer adapts their teaching, improving understanding.

Design Breakdown

Analysis of design features and their benefits

Value Proposition: LecturePulse helps lecturers understand class engagement in real time by making student focus and confusion easy to report, quick to visualise, and simple to respond to — creating more adaptive, inclusive, and focused lecture environments.

INSIGHT: Engagement within lectures is a shared responsibility, requiring both effective teaching from lecturers and a willingness from students to remain focused and actively involved in their learning.

LED Activation Light

LED provides clear feedback when the pad is active and ready for input.

Activation Button

Students place their phone on the button to activate the pad due to its weight, linking the physical interaction to the screen output.

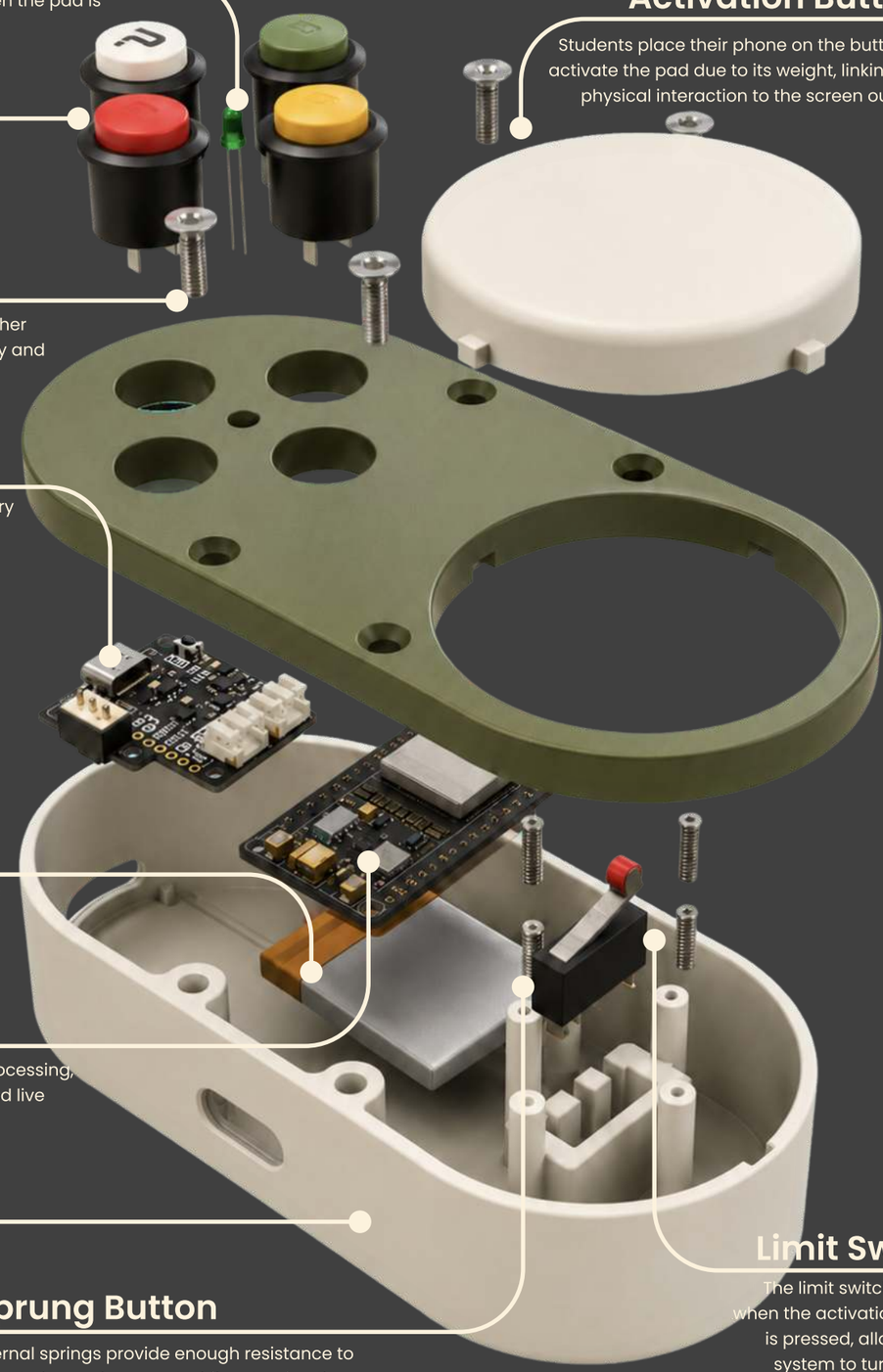
Simple 4-Button Design

Four colour-coded buttons allow students to quickly report Confused, Low Focus, Neutral Focus, or High Focus.

Lecture Integration

Works seamlessly alongside presentations, websites, and live lecture materials.





Screws

Secures the upper and lower housings together using heat-set inserts for repeated assembly and maintenance.

USB-C Charging

USB-C charging powers the internal LiPo battery through a compact power distribution board.

Confusion Indicator

When 40% of students report confusion, the lecturer receives a clear visual signal that the class may need clarification.

Internal Battery

A 3.7V LiPo battery enables portable use without requiring a fixed power connection.

ESP32 Based

An ESP32 manages input processing, wireless communication, and live feedback updates.

Focus Gradient

Students submit their current focus level, which is aggregated into a live green-to-red colour gradient on the lecturer's screen, providing an instant overview of class engagement.

Fully integrated 3d Printed Case

The 3D-printed enclosure integrates the button layout, electronics, charging port, and phone activation mechanism into a compact final form.

Sprung Button

Internal springs provide enough resistance to detect phone placement while returning the button after use.

Limit Switch

The limit switch detects when the activation button is pressed, allowing the system to turn on only when needed.

Design validation

Validating the final concept with our stakeholders

MVP goals

The MVP (minimum viable product) aimed to test whether users found the concept useful, usable and valuable enough to integrate into their learning experience.

- Real-time student feedback
- Lecturer value and usability
- Effective reaction types
- Low-distraction interaction

These 4 MVP goals were used as evaluation criteria in each design validation activity to assess whether the concept successfully met its intended objectives.

User interviews



We interviewed 15 participants to gather feedback on the updated Lecture Reaction Pad, including 5 individuals who had been involved during the initial concept selection and early prototyping stages. Participants responded positively to the revised concept, noting that previous concerns had been addressed through reducing the number of reactions to minimise distraction and introducing light gamification aspect to the game controller inspired design to encourage engagement.

☑ Real-time student feedback ☑ Low-distraction interaction ☑ Effective reaction types

We also interviewed 3 lecturers who agreed the system reduces cognitive load. Based on initial feedback from Kamyar Hazeri about avoiding added stress, the design only prompts lecturers when a 40% threshold of student responses is reached, encouraging timely but non-intrusive intervention.

☑ Lecturer value and usability

Roleplay





Roleplaying was an effective validation method as it placed the Lecture Reaction Pad within a realistic lecture context. Seeing the ambient reaction bar and question indicator on the presentation screen helped participants understand how the system would work in practice and confirmed the concept was clear and usable.

☑ Real-time student feedback ☑ Lecturer value and usability

☑ Effective reaction types ☑ Low-distraction interaction

Not done well at all — Done extremely well

1 2 3 4 5 6 7 8 9 10

Product comparison



We compared our Lecture Reaction Pad against an existing digital feedback system (Mentimeter) through repeated testing with student participants. Overall, students showed a clear preference for the physical Lecture Reaction Pad.

Key reasons included:

- reduced distraction
  - as the physical device allows phones to be put away rather than actively used
- better integration with live lecture flow,
  - since feedback is captured in real time during teaching
- increased lecturer accountability
  - as aggregated reactions are displayed on screen once a threshold is reached for the whole cohort to see

Survey results

Quick & Real-Time	7
Low Distraction in Lecture Settings	7
Easy to Interpret for Lecturers	9
Simple & Minimal Interaction	6
Encourages Student Engagement	9
Intuitive & Immediately Understandable	8
Clear Visual Feedback System	9

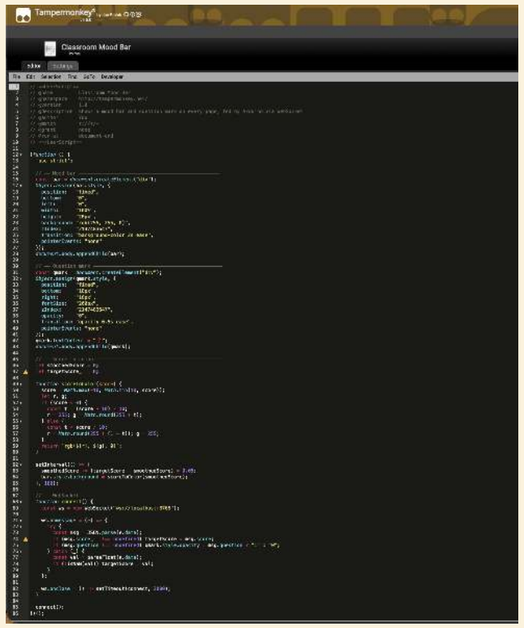
The Lecture Reaction Pad improves real-time lecture communication by reducing distraction, enabling live feedback, and increasing visible student engagement.

Appendix

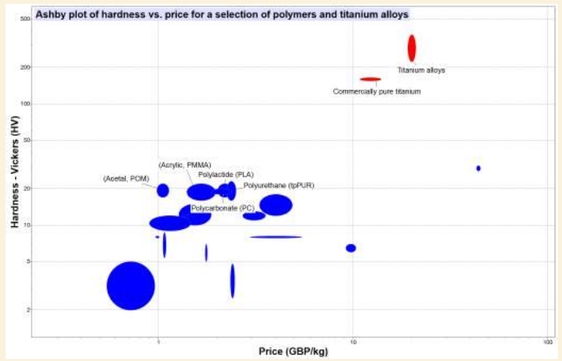
References

[1] Vinnie Wong (n.d.) *The 25 Best Shades of Red Orange (+ HEX Codes).* https://piktochart.com/blog/shades-of-red-orange/ [Accessed 18th May 2026]
[2] Alex O'Neal (2018) *Beautiful, accessible traffic light colors* https://uxdesign.cc/beautiful-accessible-traffic-light-colors-b2b14a102a38 [Accessed 3rd June 2026]
[3] Picsart (n.d) *Learn all about the color Mustard Yellow* https://picsart.com/colors/color-meanings/mustard-yellow/ [Accessed 4th June 2026]
[4] Icons8 Blog (n.d.) *Ultimate green color guide: psychology, codes, and why your brain is obsessed* https://icons8.com/blog/articles/green-color-guide/ [Accessed 18th May 2026]
[5] Ariana Ruth (n.d.) *The Science Behind Colours That Boost Focus and Calm* https://yescolours.com/blogs/news/science-behind-colours-for-focus-and-calm [Accessed 3rd June 2026]

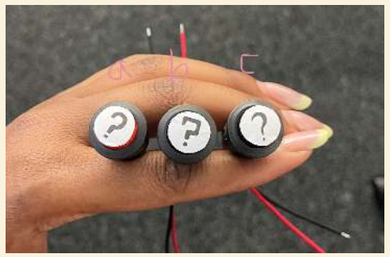
Images



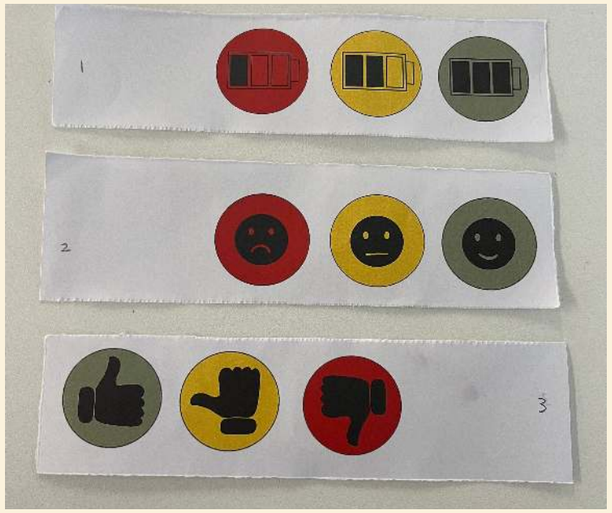
1. Code for Chrome website Visual



2. Ashby plot of polymers/titanium alloys



3. Question icon user testing props



4. Focus icon user testing paper cut outs

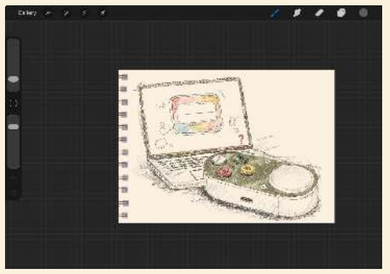


6. Focus icon user testing



5. Moodboard for lecture reaction pad CMF

Presentation design







7. Every drawing and doodle within our presentation is hand-drawn - here are examples

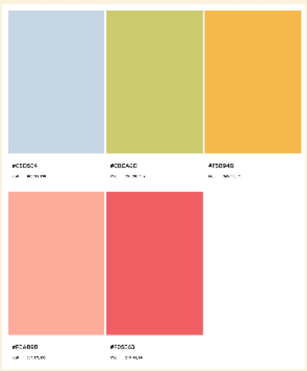
Appendix



8. CMF colour and finish

POPPINS
POPPINS
POPPINS
JUST ANOTHER
HAND

9. Fonts balance creativity and professionalism, dark grey for comfortable readability



10. Colour palette

Team collaboration

11. Figma collaboration page
https://www.notion.so/Campus-HCD-project-310f4fee4227807cac9df907531a6b44?source=copy_link

12. Notion team communication, consent forms,meeting minutes and content platform

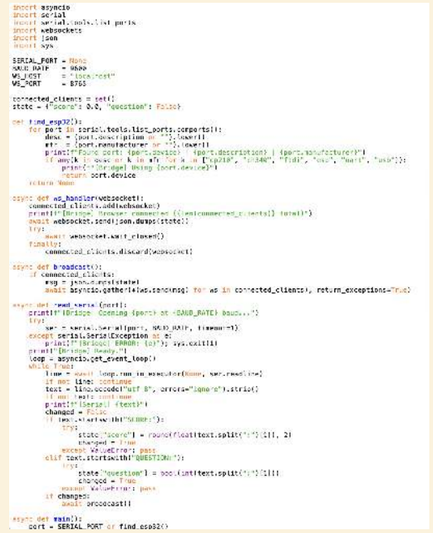
***ChatGPT was used to help edit and improve writing clarity. All research, analysis, insights, and concepts remain our own work.**

Item	Quantity	Link	Price	Delivery Date
Limit Switches	1	https://www.ama	£3.99	24 May
7mm Plastic Buttons	1	https://www.ama	£6.99	24 May
Metal Buttons	2	https://www.ama	£6.99	24 May
16mm Plastic Buttons	1	https://www.ama	£6.99	24 May
Rubber Feet V1	1	https://www.ama	£3.99	24 May
M3 Screws	1	https://www.ama	£5.92	27 May
M4 Bolts	1	amazon.co.uk/dp	£5.29	27 May
3x ESP32	1	https://www.ama	£15.19	27 May
Rubber Feet V2	1	amazon.co.uk/dp	£4.22	27 May
2mm OD Springs	1	https://www.ama	£7.49	28 May
LED Assortment	1	amazon.co.uk/dp	£5.99	28 May
100pcs Heat Inserts	1	amazon.co.uk/dp	£8.99	30 May
Micro Switch pack	1	https://www.ama	£6.99	30 May

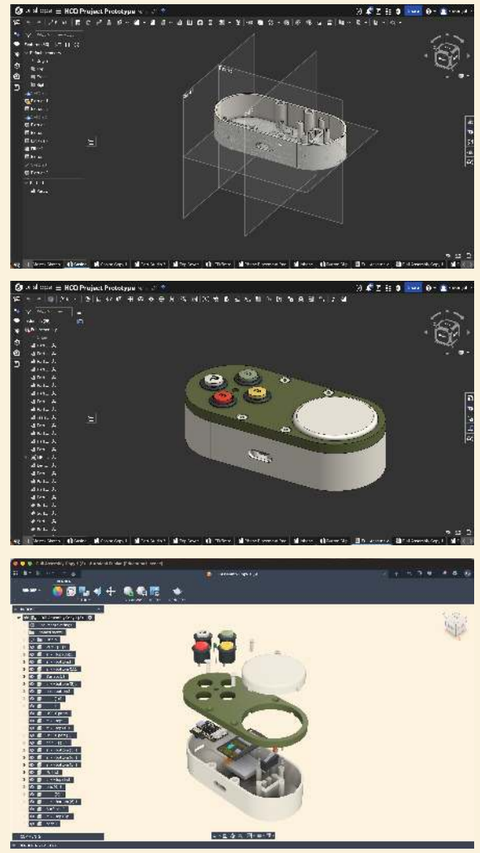
13. Parts Purchase List



15. Arduino Code



14. Serial Bridge Code



16. Assemblies in Onshape - Renders in Fusion